Contract Surplus and the Economics of External Hedging

Finite-expiry claims, qualification-aware policies, and settled-cash controls

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Abstract

A hedge changes exposure; a purchased payout right changes the payoff contract. We separate these mechanisms using an existing finite-expiry valuation and a staged synthetic experiment with independently confirmed portfolios. Under the fair law, a qualification-aware synthetic direct-account policy earns a mean $1,745 over 180 modeled calendar days, while its paired hedge increment is −$41, with a 95% Monte Carlo interval spanning zero. Reversing the orientation produces a positive contract increment but a 47.4% probability of terminal loss. An exact binary example shows why finite expiry and pre-receipt liquidity matter: a $2,000 target supports $81.22 of model surplus, whereas larger targets have negative finite-horizon value after the same fee. Acquisition price, qualification mechanics, collection, and settlement speed explain more of the economic variation than the tested hedge refinements. The contribution is a cash-based decomposition of valuable contractual rights, market exposure, and the financing needed to connect them. Results are analytical and synthetic, not estimates of realized trading income.

Keywords: contingent claims; external hedging; payout constraints; finite expiry; liquidity; Monte Carlo.
JEL classification: G13, G17, G32.

1 Introduction

An account with a loss floor and conditional withdrawal rights is not equivalent to its displayed balance. Its economic asset is the right to receive cash after surviving a sequence of barriers, qualification tests, and administrative clocks. An opposite position in an external account can offset part of the associated market exposure. It cannot, by itself, determine whether the purchased right is worth its fee. Confusing these two objects turns favorable contract economics into apparent hedge performance.

This paper asks three concrete questions. Does an external offset improve the settled-cash outcome relative to the same policy without that offset? Which account-management techniques explain the strongest defensible positive result? How much of a simple barrier-hedging argument survives a finite deadline and the need to finance losses before a receipt arrives? We answer with the completed external-hedging experiment of Bhav Jain [7], rather than a new strategy search. Its independent confirmation stage supplies the main comparison; its exact claim calculation supplies a clean analytical benchmark.

The economically useful result is a hierarchy of mechanisms. Qualification-aware targets can make a synthetic payout right more productive. An external hedge can redistribute outcomes without increasing mean value. Collection and timing can then overwhelm the surplus that remains. This hierarchy suggests evaluating the contract and the operating policy before refining the offset ratio.

2 The economic object: a right to settled cash

2.1 Account profit, claim value, and liquid wealth

Let C_t denote external spendable cash, R_t cumulative participant receipts, F_t cumulative fees, G_t gross external trading gains, and K_t external execution charges. The experiment's accounting identity is

$$C_t = C_0 + R_t - F_t + G_t - K_t, \qquad N_t = C_t - C_0. \tag{1}$$

Account ledger equity is a separate state variable. A withdrawal request is not a receipt: request reserves the amount, approval debits account equity, and only settlement credits C_t. Unpaid account equity and pending claims contribute zero to the terminal cash objective. Prop-side execution costs reduce account equity and possible payouts; subtracting them again from C_t would double-count them. This distinction is central to interpreting all subsequent tables.

There are two orientations. In *prop-primary*, the account takes positive market exposure and an external position may offset it. The comparison is the same account-management policy without that hedge. In *cash-primary*, an external positive-exposure trading program is retained and opposite prop-account positions are purchased as an overlay. The comparison is the same initial cash and external program without account purchases. "Cash-primary" is an economic role in the simulation, not evidence of live trading.

Table 1: Contract fixtures in the existing experiment. Dollar amounts are model inputs, not current purchase quotes.

Fixture	Fee	Lane cap	Economic distinction
Daily evaluation (TD)	165	5	Evaluation before funded payout rights; retained Tradeify Daily rules.
Retained MFFU EOD	209	3	Evaluation fixture with an unverified current fee and unresolved retention assumptions.
Synthetic direct EOD (DE)	492	5	No evaluation; trailing floor, retained buffer, consistency and trading-day gates.
Synthetic static ideal	492	5	Static loss floor; no retained buffer or consistency gate; an idealized comparison.

2.2 The direct-EOD fixture

The principal positive case starts with a \$2,000 loss allowance, an end-of-day trailing floor capped at \$100, and a \$1,250 session loss limit. A first payout requires \$3,000 of profit; withdrawals retain a \$2,100 buffer. Gross requests range from \$500 to \$2,000, and the participant receives 90%. Five trading days and a 20% best-day consistency condition restrict eligibility. These are *synthetic contractual terms*, not a reconstruction of Lightning or LucidDirect. The nominal \$50,000 account size is not a cash endowment.

The reference policy requests the maximum eligible amount, pauses while a request is pending, and uses approval and settlement lags of one and two modeled sessions. Purchases and replacements are financed from cash, not account equity. A floor touch is evaluated before softer daily-loss and bracket exits; a shock overshoot is realized rather than clipped. All lanes within a portfolio share the same market shock. Five lanes therefore buy five rights, not five independent market opportunities.

3 Finite expiry changes the hedge calculation

3.1 An absorption identity and its missing condition

Consider a fair binary walk in prop-account profit X, starting at zero, with lower barrier $-B$ and upper barrier T. An upper hit pays participant reward R; a lower hit pays nothing. With no expiry, the upper-hit probability is $B/(B+T)$. The claim value and a constant opposite hedge magnitude are

$$V_\infty = \frac{RB}{B+T}, \qquad h = \frac{R}{B+T}. \tag{2}$$

At either absorbing endpoint, reward plus the gain on a short h units equals V_∞. After purchase fee f, both endpoint outcomes equal $V_\infty - f$. But an unhit expiring claim has no reward: its combined outcome is $-hX_H - f$, not generally $V_\infty - f$. The static identity has not insured the third outcome.

For the separate finite contract, let $V_n(x)$ be the continuation value before event $n+1$, and let $A_\pm$ be the immediate receipt plus continuation value after the next $\pm\delta$ shock. Binary replication gives

$$V_n(x) = \tfrac{1}{2}(A_+ + A_-), \qquad q_n(x) = -\frac{A_+ - A_-}{2\delta}. \tag{3}$$

A last-event barrier hit settles before expiry; unhit continuation is zero. The negative sign makes q a hedge *of* the purchased claim rather than a position replicating ownership of it. Recursive cancellation then fixes terminal receipts plus hedge gains minus the fee at $V_0 - f$ on every binary path.

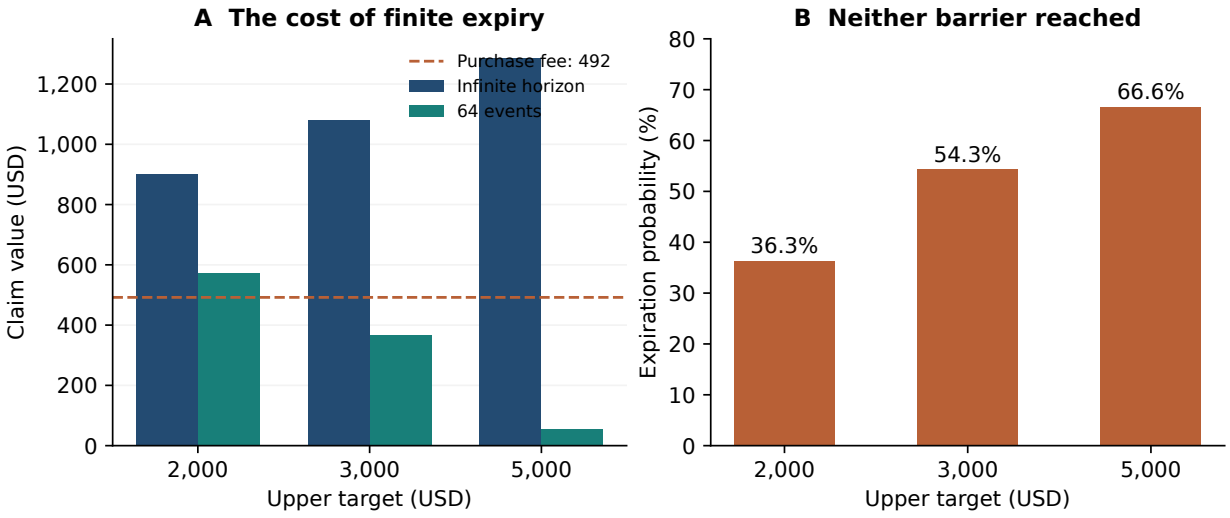


Figure 1: Existing exact-rational results: $B = \$2,000$, $R = 0.9T$, $f = \$492$, $\delta = \$250$, and 64 events. A larger advertised target raises the infinite-horizon value while lowering the finite value in these three declared cases. Bars show analytical values, not Monte Carlo estimates.

Table 2: Exact finite-claim valuation and worst required initial cash, USD.

Target	Finite value	Net	Infinite value	Cash need	Expiry
2,000	573.22	81.22	900.00	1,718.78	36.31%
3,000	366.15	-125.85	1,080.00	2,825.85	54.34%
5,000	56.24	-435.76	1,285.71	4,935.76	66.64%

The \$2,000 target retains a positive \$81.22 surplus, but financing the hedge before receipt requires \$1,718.78 of initial spendable cash. At \$5,000, the net is −\$435.76 and cash need is \$4,935.76. These are properties of a binary, fractional, frictionless, immediate-settlement claim; the lifecycle experiment below has different payout gates and is not covered by this pathwise identity.

4 Experimental design and the relevant control

4.1 A staged comparison, not a historical backtest

The existing study contains 590 distinct specifications including controls and exact claims. Screening uses 256 independent portfolios per job; resource selection also uses 256, with a separate seed. The 16 resource-selected specifications are then confirmed on 2,048 independent portfolios each. Stress uses 1,024 portfolios per job and a fourth seed. Complete stages contain 380, 151, 29, and 104 jobs respectively; controls explain why job counts exceed selected-arm counts. Neither confirmation nor stress is used to rerank policies.

Each portfolio spans 180 synthetic calendar days starting on a Monday. Weekends consume random draws but do not trade; there is no historical holiday calendar. A weekday contains 16 shocks of \$200, giving an unstopped fair-law daily standard deviation of \$800 for one unit. The two reference laws, 0.50 and 0.55, describe eventual symmetric \$1,000 bracket success, not per-shock success or observed skill. With $m = 1000/\delta$, the per-shock probability is

$$p = \left[1 + \exp\left\{-\frac{1}{m}\log\frac{w}{1-w}\right\}\right]^{-1}. \tag{4}$$

The 0.55 law favors the positive primary direction. Opposing exposures do not both receive that advantage. This is a controlled directional scenario, not a forecast fitted to market data.

4.2 Paired increments identify the hedge contribution

Common random numbers couple each candidate and its control within a stage. For simulated portfolio i, write $D_i = N_i^{\text{candidate}} - N_i^{\text{control}}$. The reported interval is

$$\bar{D} \;\pm\; 1.96\frac{s_D}{\sqrt{n}}. \tag{5}$$

It uses the dispersion of paired differences, not two independent-arm standard errors. The interval describes Monte Carlo integration uncertainty conditional on the model. Policy selection precedes confirmation; this is out-of-sample simulation, not out-of-sample market evidence [4].

Table 3: Independent confirmation: 180-day settled net and the matched-control increment, USD. TD is the daily-evaluation fixture; DE is synthetic direct EOD. All eight TD/DE product–orientation–law groups are shown.

Fixture	Primary	Law	Total net	Increment	Paired 95% CI
TD	Prop	0.50	-460.33	526.54	[-843.41, 1,896.49]
TD	Cash	0.50	-1,172.55	-1,087.79	[-1,576.49, -599.08]
DE	Prop	0.50	1,745.40	-40.97	[-677.85, 595.91]
DE	Cash	0.50	2,464.59	2,549.35	[1,895.93, 3,202.77]
TD	Prop	0.55	4,681.93	-1,686.11	[-1,802.84, -1,569.37]
TD	Cash	0.55	7,206.40	-951.02	[-1,008.90, -893.14]
DE	Prop	0.55	7,295.51	0.00	[0.00, 0.00]
DE	Cash	0.55	6,502.44	-1,286.62	[-1,460.94, -1,112.30]

In prop-primary, a positive total with a near-zero increment means the purchased rights and management policy, not necessarily the hedge, explain the result. In cash-primary, the increment measures the purchased overlay plus any change in the external program induced by cash constraints. That second channel must not be described as the insurance payoff alone.

5 What survives independent confirmation?

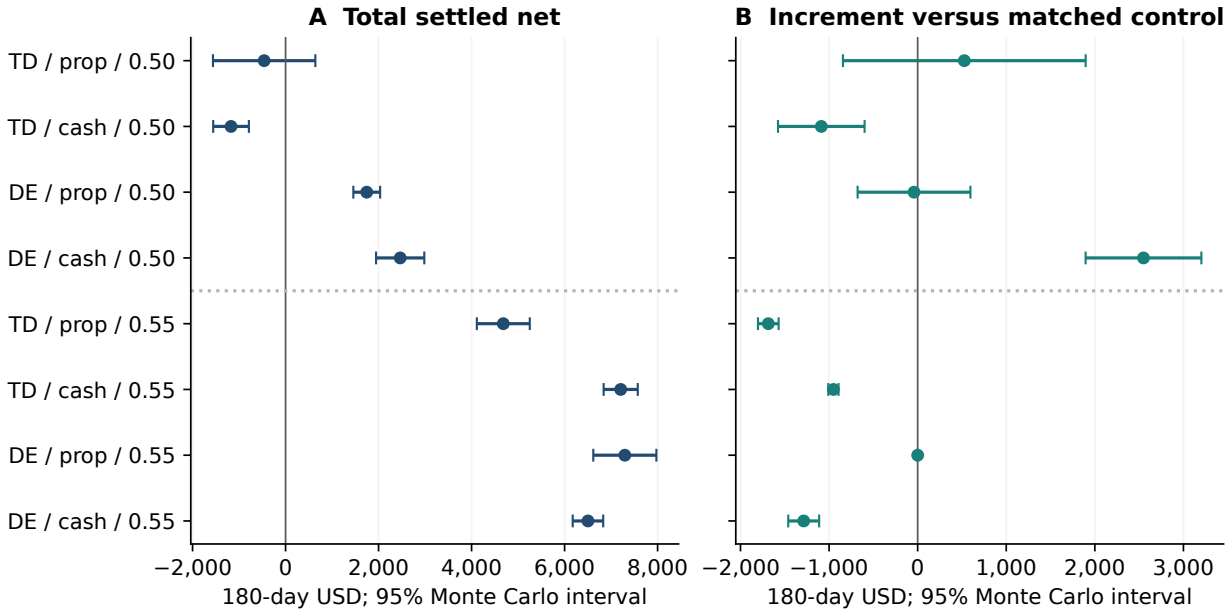


Figure 2: A profitable combined program need not contain a profitable hedge. Dots and bars are means and 95% Monte Carlo intervals, not terminal outcome ranges. The dashed horizontal separator distinguishes the fair and directional laws. The DE prop-primary 0.55 increment is exactly zero because its selected policy is unhedged.

5.1 A positive direct-account business, a neutral hedge increment

The fair-law DE prop-primary selection uses five lanes, $25,000 initial cash, qualification-aware targets, $1,000 risk limits, and a static half hedge. Its mean settled net is $1,745.40, with a

total-mean interval of [$1,457.49, $2,033.31]. Its same-policy unhedged control earns $1,786.37. The hedge increment is therefore −$40.97, with interval [−$677.85, $595.91]. This supports a positive modeled contract-and-policy combination, not an incremental mean-return claim for the offset.

The fair-law cash-primary selection uses five opposite qualification-aware lanes and only $10,000 initial cash, while retaining one positive external unit. Its $2,464.59 total and $2,549.35 increment are positive, but the loss probability is 47.41%. The paired external-gross difference includes approximately $61.87 of altered realized base trading caused by cash constraints. Thus even a positive overlay increment should be decomposed before it is labeled insurance value.

5.2 Why the other controls matter

Neither named evaluation fixture establishes fair-law profitability. TD prop-primary averages −$460.33; TD cash-primary averages $-\$1,172.55$. The retained MFFU counterparts average −$825.99 and −$662.39. Conversely, the static-ideal contract produces $62,384.79 in fair-law prop-primary mean net, yet its hedge increment is only −$45.41, with an interval spanning zero. Removing contractual friction creates valuable rights; it does not demonstrate a valuable hedge technique or an available product.

Under the 0.55 law, TD and DE cash-primary total means become positive while both overlay increments are negative. These policies purchase offsetting exposure against an advantaged primary direction. Comparing only combined profits would conceal the price of that insurance. The relevant question is whether the altered risk distribution justifies its incremental cost, not whether the advantaged base program remains profitable after paying it.

6 The stronger mechanism: qualification-aware targets

An important naming distinction is that the existing "qualification sizing" policy changes the session profit target, not the number of units or the $1,000 loss bracket. It makes the objective responsive to payout eligibility. Let E be current equity, E_c cycle-start equity, G the required eligibility equity, d remaining qualifying days (at least one), c the consistency fraction, and r the configured risk. For the direct fixture's trading-day gate, the implementation's target, in integer cents, reduces to

$$u = \max\{1, \lceil (G - E)/d \rceil\}, \qquad a = \min\{r, \max[1, \min(u, \lfloor c \max(1, G - E_c) \rfloor)]\}. \tag{6}$$

The eligibility level G jointly covers the retained buffer and minimum request, one-cent clearance above the post-withdrawal floor, first-cycle profit, and best-day consistency. The actual engine also covers winning-day and evaluation gates; equation (6) describes the focal synthetic direct fixture, where the winning-day profit threshold is zero. Exits occur after the full shock, so the target is not an exact fill-price promise.

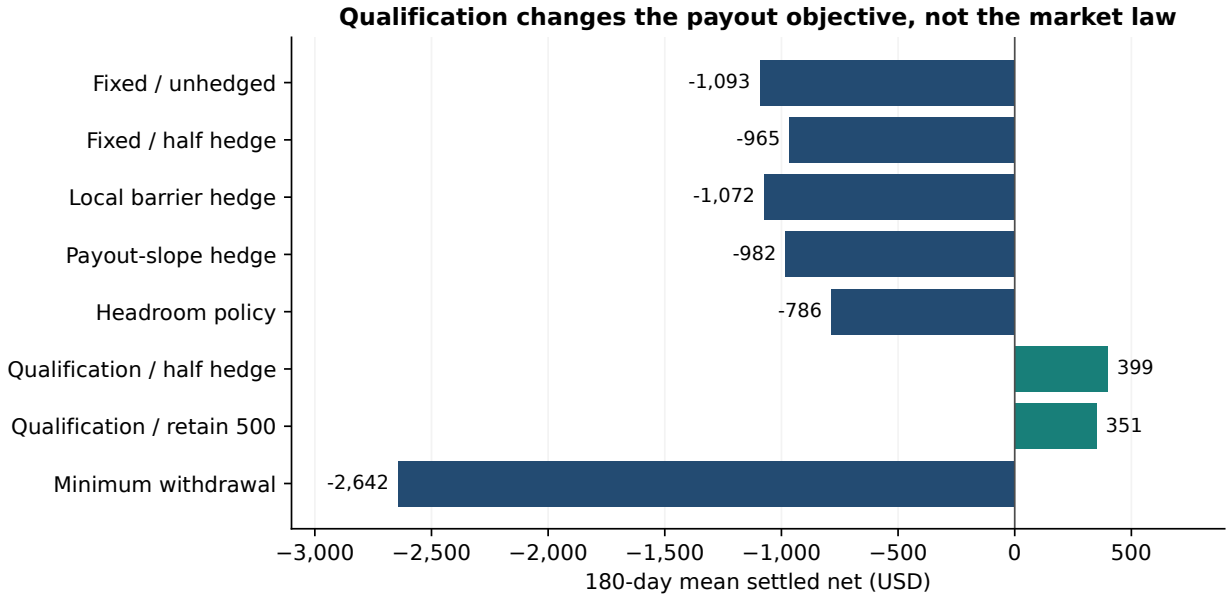


Figure 3: Existing fair-law screen: one DE lane, \$10,000 cash, \$1,000 risk, 256 portfolios per arm. The displayed menu is a mechanism comparison, not a new optimization. Qualification, room, minimum-withdrawal, and retention variants use the configured half hedge unless otherwise labeled. These are screen means; confirmation of the selected scaled policy is reported separately.

Fixed sizing without a hedge loses \$1,092.84 in this screen. Local barrier and payout-slope hedges also remain negative, at $-\$1,072.09$ and $-\$981.75$. Qualification-aware targets with a half hedge instead earn \$398.91; their same-policy unhedged control earns \$437.98. The improvement is therefore more consistent with better conversion of account states into payout access than with an offset generating edge. An additional \$500 retention requirement reduces the qualification result to \$351.13. Minimum withdrawals perform particularly poorly, at $-\$2,641.61$.

These are not isolated randomized treatment effects: the displayed menu changes different policy features and has finite screening noise. Its value is mechanistic. Stopping after a useful day's profit can preserve the consistency of the payout cycle; minimizing every cash request can instead delay recovery of acquisition fees. "More conservative" at the account level is not synonymous with a higher value of collected cash.

7 Price matters more than a finely tuned static offset

The resource stage contains a predeclared 6×5 fee–hedge grid for the synthetic direct-EOD fixture. It holds a baseline fixed-target policy, one lane, \$10,000 cash, and the fair law constant. The acquisition fee varies over \$100, \$200, \$300, \$492, \$700, and \$1,000; the static hedge fraction varies over 0, 0.25, 0.50, 0.75, and 1. These nodes are descriptive and were not eligible for selection. Crucially, this is *not* the qualification-aware policy whose scaled result appears in confirmation.

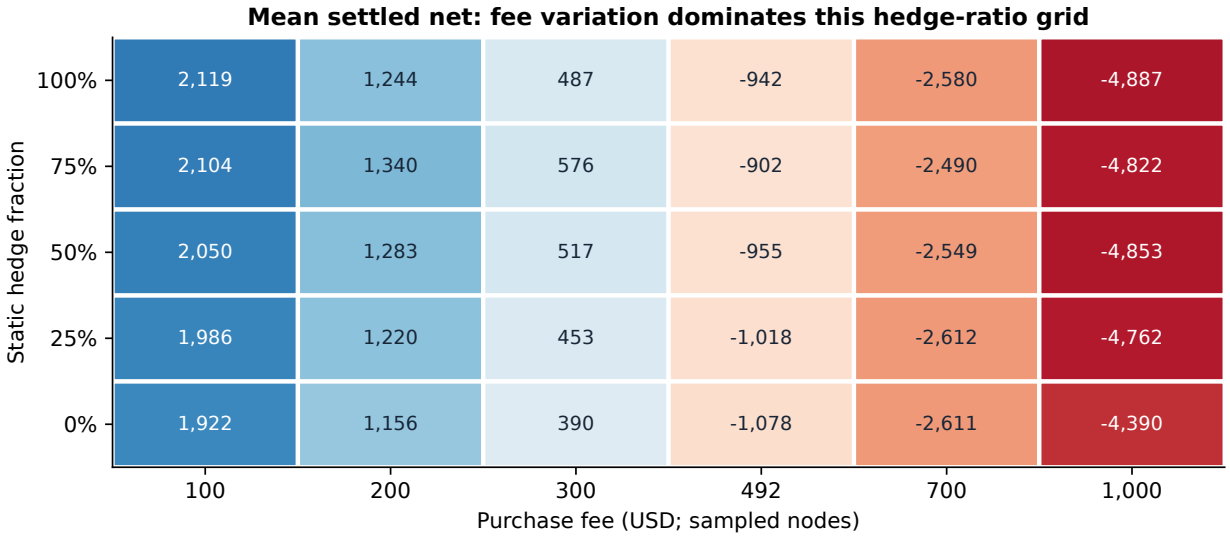


Figure 4: Frozen price–hedge nodes: 180-day mean settled net in USD, based on 256 portfolios per node. Each rectangle represents one declared parameter pair; column widths are categorical, and no interpolation or fitted break-even price is implied. Blue is positive and red negative. Fees are illustrative prices, not verified offers for the synthetic rights.

Without a hedge, the mean is \$1,922.34 at a \$100 fee, \$389.53 at \$300, and $-\$1,078.13$ at \$492. Every tested fraction has a positive sample mean at \$300 and a negative sample mean at \$492. This brackets a sampled sign change; it does not solve for a continuous break-even offer. At \$492, a 0.75 fraction raises the mean to $-\$901.94$, still negative. Lowering the acquisition price moves the account business across zero more decisively than those local hedge adjustments.

A simple decomposition explains the sensitivity. With M purchases,

$$N_H = R_H - fM + G_H - K_H. \tag{7}$$

If purchases and subsequent decisions were held fixed, a dollar fee reduction would add M dollars to net. The actual finite-cash model can also change M and the timing of trades, so this is an accounting explanation rather than a constant derivative estimated from the surface. Account acquisition is repeated, which amplifies a modest price difference over a 180-day experiment.

Scale has a similarly specific meaning. In the existing fair-law resource stage, the selected qualification mechanism yields \$304.95 with one lane and \$1,524.73 with five lanes and \$25,000 cash; staggering the latter gives \$1,463.09. These are resource-stage means, not the confirmation estimates. Adding lanes increases access to the same rights on the same market tape. It does not diversify the shock driving them, and cannot be interpreted as an independent-account law of large numbers.

8 Collection and settlement are first-order economic inputs

A payout right depends on more than whether a price barrier is crossed. To separate sampling variation from scenario effects, the study reruns each frozen selection and its control on the common stress-stage seed. Table 4 compares these stress results with the *stress reference*, not with a confirmation estimate from a different random sample. All rows retain the fair market law and the existing selected policy, except for the stated change.

Table 4: Fair-law DE stress results: total and paired incremental 180-day mean net, USD; 1,024 portfolios per job. Each increment uses its own matched control. "Cash" means the cash-primary orientation.

Scenario	Prop net	Prop incr.	Cash net	Cash incr.
Reference	2,059.17	-230.88	2,408.79	2,849.42
Unit charge: 2	1,418.74	-1,455.30	2,552.67	3,459.93
Collection: 90%	-58.64	-278.36	565.16	1,005.79
Approval/settlement: 2/5	181.56	-84.34	595.38	1,036.00
64 shocks of 100	70.47	40.75	970.24	1,028.15
Trade pending; forfeit	2,316.19	-3.14	2,608.17	3,048.80
Three-request stop	-8,818.95	-213.75	-7,594.01	-7,153.39
Initial cash: 5,000	1,374.91	173.27	1,950.40	2,396.10

The prop-primary reference earns \$2,059.17. Reducing collection probability to 90% changes the mean to −\$58.64; slowing approval/settlement lags from 1/2 to 2/5 sessions leaves \$181.56. Both stressed total-mean intervals include zero. The corresponding cash-primary means fall from \$2,408.79 to \$565.16 and \$595.38. Their overlay increments remain positive, but their total-mean intervals also include zero. Collection affects realized cash; delay additionally parks lanes and reduces the number of cycles completed before the horizon.

The \$2 one-way unit execution-charge scenario leaves prop-primary total net positive at \$1,418.74, but its hedge increment falls to $-\$1,455.30$, with paired interval $[-\$2,424.66, -\$485.94]$. A total-only report would miss this cost of the hedge. These charges are a synthetic stress rather than a broker commission schedule; margin, financing, and taxes are not included.

The finer grid replaces sixteen \$200 shocks with sixty-four \$100 shocks. Both have unstopped fair daily variance 640,000, yet the prop-primary mean falls to \$70.47. Equal variance does not imply equal stopping, overshoot, or qualification behavior. This sensitivity is especially relevant to the small profit targets in equation (6); the grid is part of the model, not an innocuous numerical plotting choice.

Finally, the synthetic three-request firm-stop scenario produces large losses in both orientations. It is not an official firm rule or an estimated termination hazard. Its role is to expose dependence on continued access and repeated collection. Conversely, allowing trading while pending can improve throughput even when approved claims may be forfeited after a breach. Pausing all activity is not automatically economically conservative: it protects one state while consuming the finite time available to create others.

9 Positive average value is not a smooth cash stream

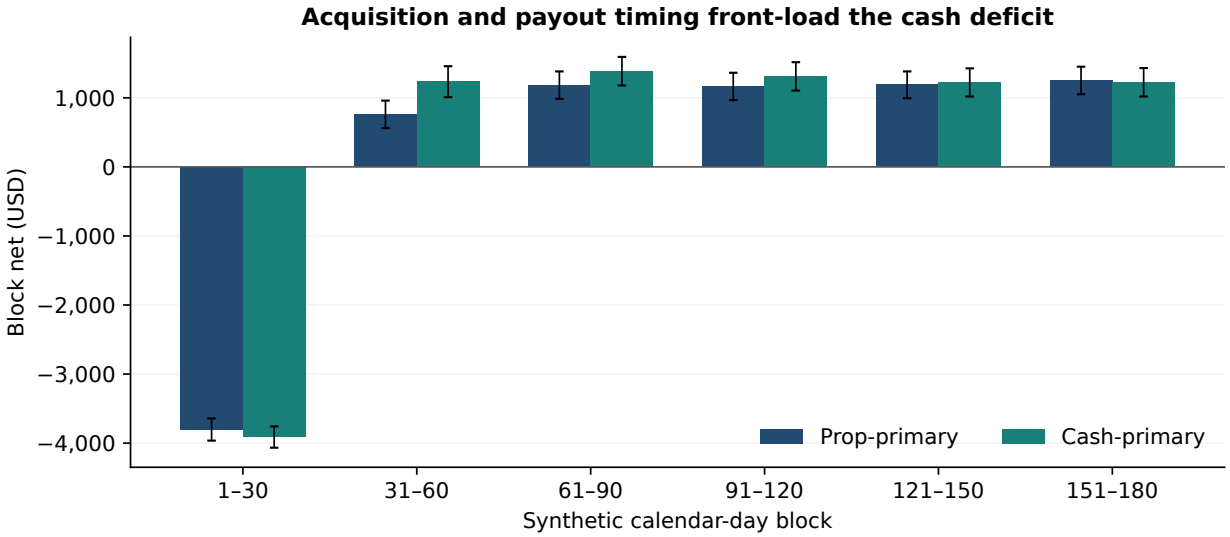


Figure 5: The six actual 30-calendar-day blocks of the fair-law DE confirmation. Bars are block means; whiskers are Monte Carlo intervals for those means. The initial block includes acquisition and qualification costs before sustained collections. These are synthetic blocks, not historical L30-session windows.

Table 5: Fair-law DE confirmation: terminal risk and intraperiod cash pressure. Dollar rows are USD. The two selected policies have different initial wallets.

Measure	Prop-primary	Cash-primary
Initial cash	25,000.00	10,000.00
Mean net	1,745.40	2,464.59
Median net	1,580.00	705.00
5th percentile net	-8,943.00	-9,999.98
Worst-5% mean net	-11,589.55	-9,999.98
Probability of loss	40.62%	47.41%
Any cash clipping	0.00%	54.88%
95th percentile cash drawdown	15,790.00	14,629.99

The cash-primary arithmetic is particularly revealing: \$17,978.99 of receipts minus \$15,491.51 of fees and \$22.89 of gross external loss yields \$2,464.59 net. A comparatively small surplus sits between much larger gross flows. Its first block loses \$3,912.32 on average. Dividing the full-horizon mean by six would yield \$410.76, but that number does not describe the cash experience of starting the program.

The 47.41% terminal loss probability, \$705 median, and approximately $-\$10,000$ lower tail make the distinction sharper. Cash clipping occurs in 54.88% of simulated portfolios. The reported worst-5% mean is a tail average, not a guaranteed loss limit; this risk concept is distinct from a confidence interval for the mean [6]. Cash solvency is modeled at intraperiod events, but a spendable-cash calculation is not a broker-margin requirement.

10 Interpretation and contribution

10.1 What claim theory explains—and what the lifecycle adds

Binary replication makes the source of a locked-in surplus explicit: the purchased right costs less than its model value. The dynamic hedge transfers state-contingent payoff into a constant within that complete model; it does not raise the fair expected gross payoff. This is the same distinction between valuation and physical return that underlies contingent-claim pricing [1, 2, 3]. Under predictable integrable exposure and a finite fair horizon, the external gain has zero expectation:

$$\mathbb{E}\left[\sum_{n=0}^{H-1} q_n \Delta X_{n+1}\right] = \sum_{n=0}^{H-1} \mathbb{E}[q_n \mathbb{E}(\Delta X_{n+1} \mid \mathcal{F}_n)] = 0. \tag{8}$$

Finite-cash constraints can nevertheless make a hedge change acquisition, activity, and ultimately receipt opportunities. This explains why a lifecycle increment need not be identically zero even in a fair law. In the focal prop-primary confirmation, however, the estimated increment does not establish a positive mean benefit.

The lifecycle also makes the liquidity problem concrete. A lower terminal loss frequency can coexist with expensive interim financing, delayed receipts, and lost trading opportunities. This connection between risk management and the ability to fund valuable activities is familiar in corporate hedging theory [5]. Here it appears through account fees and payout clocks rather than physical investment. A market offset cannot insure denial of an otherwise valid claim, loss of account access, or the continued availability of favorable terms.

10.2 Scope of inference

These are research-only results under conditional synthetic laws, not empirical return calibration. Tradeify's audited rules prohibit the modeled opposite positions, and the synthetic DE fixture is not an offered product. MFFU pricing and retention remain unresolved in its retained scenario. The local barrier and payout-slope rules are heuristics, not solutions to the joint lifecycle hedge. The exact certificate has neither those lifecycle rules nor settlement delay. These distinctions delimit the meaning of the numerical results without removing their comparative content.

10.3 Conclusion

The strongest defensible result is not a universal hedge but an ordered economic analysis. First price the collectable right, including its finite deadline. Then determine how trading objectives convert account states into eligible withdrawals. Finally measure the external overlay against a cash-matched control and inspect the financing path. In this experiment, qualification-aware targets support a positive synthetic direct-account result, while the tested hedge does not establish additional mean value. Fee levels, collection, and timing move that result more than local offset refinements. The exact calculation explains why: a hedge can organize the payoff of a valuable right, but cannot make an unvaluable expiring right valuable merely by matching its absorbing endpoints.

Data and materials

The accompanying research materials provide the manuscript source, aggregate results, and vector figures [8]. Numerical results come from the completed September 2026 external-hedging study [7]; figures are redrawn from its confirmation, screening, resource, stress, and exact-valuation

outputs. The companion `aggregates.json` preserves numerical precision, source references, and candidate–control mappings rather than individual simulated paths.[1]

This separation is intentional. Screening results in Figure 3 explain a mechanism; independent confirmation in Table 3 evaluates the already selected policy; the finite-claim values in Table 2 explain an analytical identity. They are not interchangeable estimates of the same object. No two experiments' cash flows are added together.

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[1] Historical dates are N/A: no market sessions were admitted. L5/L10/L30 retain empty date lists and daily rows, with null cutoffs and economics; synthetic calendar-day blocks are reported separately.